Him-Rakshak

AI-Based Early Warning & Landslide Risk Monitoring System for North Eastern Region (NER)

Problem Statement ID	SIH26001
Organization	Ministry of Development of North Eastern Region (MDoNER)
Category	Software
Theme	Disaster Management
Event	Smart India Hackathon (SIH) 2026
Team Size	6 Members

1. Background

The North Eastern Region (NER) frequently faces landslides, flash floods, road blockages, and slope failures due to heavy rainfall, fragile terrain, and unplanned hill cutting. Monitoring is currently reactive and dependent on manual reporting, with limited real-time predictive capability. Him-Rakshak proposes an AI-enabled real-time monitoring and prediction platform to help authorities take preventive action before disasters occur.

2. Problem Statement Requirements

The solution is expected to:

- Collect and analyse rainfall, soil moisture, satellite imagery, terrain/slope, and historical landslide data
- Use AI/ML models to identify high-risk zones and predict landslide events
- Provide real-time alerts to district administrations, disaster authorities, and communities
- Integrate GIS mapping for vulnerable roads, villages, and infrastructure
- Allow citizens/field officials to upload geo-tagged photos/videos of cracks or blocked roads
- Generate dashboards for risk severity, road connectivity, weather-linked forecasts, and emergency prioritisation
- Support multilingual notifications and low-network/offline functionality

3. Proposed Solution – Him-Rakshak

Him-Rakshak is a full-stack AI-powered platform combining real government data sources, a machine learning risk-prediction engine, a live GIS dashboard, a citizen field-reporting portal, and an automated multilingual alert system.

System Architecture (data flow)

IMD Weather API + GSI Historical Records + Bhuvan/SRTM Terrain Data → Data Pipeline → ML Risk Prediction Model (Random Forest / XGBoost) → FastAPI Backend + PostgreSQL Database → React + Leaflet GIS Dashboard → Multilingual Alert Engine (SMS / App notifications). Citizens and field officials feed geo-tagged photo/video reports back into the system through a web form.

4. Technology Stack

Layer	Technology
Machine Learning	Python, pandas, scikit-learn, XGBoost
Backend / API	FastAPI (Python)
Database	PostgreSQL (via Supabase)
Frontend / Dashboard	React, Leaflet.js, Tailwind CSS
Alerts	Twilio (SMS simulation), multilingual templates
External Data APIs	IMD Weather API, GSI Bhukosh, ISRO Bhuvan / SRTM DEM
Deployment	Vercel (frontend), Render/Railway (backend), Supabase (DB)
Version Control	Git + GitHub

5. Real Data Sources Used

- **Rainfall / Weather:** India Meteorological Department (IMD) public API – live district rainfall, warnings, and nowcasts
- **Historical Landslides:** Geological Survey of India (GSI) – Bhukosh portal
- **Terrain / Slope:** ISRO Bhuvan and SRTM Digital Elevation Model (DEM)
- **Satellite Imagery:** ISRO Bhuvan / Sentinel-2 (sample imagery for prototype)
- **Soil Moisture:** Simulated / proxy values (real IoT sensor deployment is future scope)

6. Machine Learning Approach

- Rainfall-threshold baseline model adapted from established intensity-duration formulas for Himalayan/NER terrain
- Random Forest / XGBoost classifier to categorise zones as Low / Medium / High / Critical risk
- Features: rainfall intensity, slope angle, soil-moisture proxy, and historical landslide frequency
- Model served via a FastAPI endpoint that returns a risk score for a given location

7. Requirement Coverage

Honest mapping of the prototype against the official expected solution:

Requirement	Status
Rainfall pattern analysis	Fully covered (live IMD API)
Terrain / slope data	Fully covered (SRTM / Bhuvan)
Historical landslide records	Fully covered (GSI Bhukosh)
AI/ML risk-zone prediction	Fully covered
GIS mapping (roads, villages, infra)	Fully covered (React + Leaflet)
Citizen / field photo-video reporting	Fully covered (web form)
Dashboards (risk, roads, forecast, priority)	Fully covered
Multilingual notifications	Fully covered (English / Hindi / Assamese)
Soil moisture sensors	Simulated (real IoT hardware is future scope)
Live satellite feed	Sample imagery only (satellites revisit every few days)
Real-time SMS delivery	Simulated for demo (production needs telecom integration)
Offline sync support	Designed conceptually; not fully implemented in prototype

Note: The partially-covered items require physical IoT hardware, satellite revisit cycles, or production-grade telecom/cloud infrastructure that are outside the scope of a short hackathon build timeline. These are addressed in the Future Scope section below.

8. Development Timeline

Day	Focus
Day 1	Team roles, repo setup, data collection (IMD, GSI, Bhuvan)
Day 2	ML model training, backend skeleton
Day 3	Backend API + Dashboard (risk heatmap, road status) + reporting form
Day 4	Dashboard (forecast + emergency priority panels), alerts, multilingual support
Day 5	Integration, testing, PPT, demo video, rehearsal

9. Team Roles (6 Members)

- Member 1: Data collection & ML model training
- Member 2: Backend API development
- Members 3 & 4: Dashboard / frontend development
- Member 5: Field-reporting form & alert system
- Member 6: Documentation, PPT, and demo video

10. Future Scope

- Deployment of real IoT sensor networks (soil moisture, tiltmeters, vibration sensors) on vulnerable slopes
- Fully offline-first mobile application with local caching and automatic sync
- Automated satellite change-detection pipeline using Sentinel-2 / Bhuvan imagery and computer vision
- Improved prediction accuracy using deep learning (LSTM) on larger historical datasets
- Scale-up to all seven NER states with district administration integration
- Integration with GSI's National Landslide Forecasting Centre and NDMA systems
- Production-grade automated SMS/IVR alert gateway for remote and low-literacy areas
- Community-driven crowdsourced verification network for ground-truth data

11. Expected Impact

Him-Rakshak aims to shift landslide management in the North Eastern Region from a reactive to a predictive model — improving disaster preparedness, reducing loss of life and infrastructure damage, and strengthening climate-resilient governance in one of India's most vulnerable regions.

Prepared as a project overview document for Smart India Hackathon 2026 (SIH26001).